

SHUBHAM MASAL

Robotics & Automation Engineer |
Aspiring Data Scientist & ML Engineer

ABOUT ME

Automation & Robotics Engineer transitioning into Machine Learning Engineering, with hands-on industrial experience and advanced data science expertise. Proven track record in building end-to-end data pipelines, with practical experience deploying supervised and unsupervised ML models, computer vision pipelines (object detection and image processing), predictive analytics frameworks & leveraging LLMs for Gen AI applications. Proficient in Python, ML/DL frameworks, and SQL, backed by a strong foundational track record in industrial automation systems, hardware-software integration, and complex systems troubleshooting. Eager to leverage this unique blend of industrial automation and data-driven intelligence to build robust, scalable ML pipelines and AI-driven solutions.

PROJECTS

DS/ML PROJECTS

AERIAL OBJECT CLASSIFICATION & DETECTION

| Python, Computer Vision, Transfer Learning, YOLOv8, Streamlit

Built a real-time system to classify and localize Birds vs. Drones. Selected ResNet50 after achieving 98.6% accuracy and 1.00 Precision (0 false positives) against a custom CNN. Integrated YOLOv8 and a robust preprocessing pipeline to handle lighting and silhouette ambiguity. Launched the final optimized model as an interactive web application.

CROSS MARKET ANALYSIS

| Python, MySQL, SQLAlchemy, Streamlit, Pandas, REST APIs

Built a Python/SQLAlchemy pipeline ingesting CoinGecko, Yahoo Finance, and GitHub data. Designed a relational schema integrating Crypto, WTI Oil, and global stock indices. Wrote 30+ complex SQL queries to analyze asset correlations and macroeconomic impacts. Developed an interactive web UI for asset visualization.

DIAMOND DYNAMICS: PRICE PREDICTION & MARKET SEGMENTATION

| Python, EDA, ANN, Supervised & Unsupervised Algorithms, Streamlit

Engineered a portfolio-grade system combining supervised regression for valuation and unsupervised clustering for asset segmentation. Evaluated multiple models, selecting an optimized XGBoost Regressor (R^2 : 0.9815, MAE: 274.75) that outperformed a PyTorch ANN (R^2 : 0.9785). Applied PCA to compress physical attributes (61.6% variance captured), feeding into a DBSCAN clustering pipeline to establish pricing bands.

EDUCATION REGULATION IMPACT ANALYZER (ERIA)

| Python, NLP, Groq API, Hugging Face (BART), Streamlit, AWS

Built a GenAI system converting raw policy PDFs into structured compliance insights. Used Hugging Face (BART) for zero-shot classification and the Groq API for summarization and risk forecasting. Developed a modular, config-driven repository featuring automated preprocessing and structured JSON outputs. Hosted the interactive application on the cloud for real-time text extraction and visualization.

EDUCATION

SIU'S SYMBIOSIS INSTITUTE OF TECHNOLOGY, PUNE

Master's in Automation & Robotics
2023 - 2025

SAVITRIBAI PHULE PUNE UNIVERSITY, DPCOE, PUNE

Bachelor's in Mechanical Engineering
2020 - 2023

STES'S SOU. VENUTAI CHAVAN POLYTECHNIC, PUNE

Diploma in Mechanical Engineering
2017 - 2020

CERTIFICATIONS

- HCL GUVI Master Data Science Program
- EDX VERIFIED DATA SCIENCE WITH PYTHON
- Udemy's Machine Learning A-Z
- EDX Engineering Design for Circular Economy
- NPTEL Certification – Fundamentals of Artificial Intelligence
- NPTEL Certification – Robotics
- NPTEL Certification – Foundation of Cognitive Robotics
- Elevation's ANSYS Basic To Professional Training
- AUTODESK AutoCAD 2015

CONTACT

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🌐 <https://github.com/smasal23>

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FOREST COVER TYPE PREDICTION

| Python, Scikit-Learn, Pandas, EDA, Feature Engineering, Streamlit

Engineered a reproducible, config-driven ML workflow to predict 7 forest cover types using 20 terrain features. Handled an extreme 47.72 class imbalance across 145K observations by optimizing for Macro F1-Score. Created domain-specific spatial interaction features (e.g., elevation-slope) with strict data-leakage safeguards. Evaluated 8 algorithms via 5-fold cross-validation, selecting and tuning a Random Forest architecture.

GLOBAL LITERACY & EDUCATIONAL TRENDS ANALYSIS

| Python, MySQL, Power BI, Pandas, Insight Presentation, REST APIs

Built a Python pipeline to retrieve, audit, and clean heterogeneous OWID socio-economic datasets. Architected a multi-table database using composite natural keys to run 30+ cross-market analytical SQL joins. Built a multi-page PowerBI dashboard featuring drill-through logic and decomposition trees to translate SQL outputs into stakeholder insights.

ADDITIONAL AUTOMATION ENGINEERING PROJECTS

REFURBISHMENT, TRANSITION & INTEGRATION OF AUTOMATIC FOIL CAPACITOR WINDING MACHINE(MTECH-INTERNSHIP PROJECT)

DESIGN & DEVELOPMENT OF GLASS TILTING MECHANISM FOR AUTOMATING GLASS HANDLING(MTECH-INTERNSHIP PROJECT)

AUTONOMOUS FIRE SAFETY RESPONSE SYSTEM (MINI PROJECT)

REAL-TIME NEWS UPDATE BOT WITH UIPATH (MINI PROJECT)

SMART AUTOMATIC DOOR SYSTEM (MINI PROJECT)

DESIGN & FABRICATION OF REGENERATIVE BRAKING SYSTEM (GRADUATION PROJECT)

CENTRAL TYRE INFLATION SYSTEM (DIPLOMA PROJECT)

TECHNICAL SKILLS

PROGRAMMING & DATA SCIENCE

- Python
- SQL
- Pandas, Numpy, Matplotlib, Seaborn
- Scikit-Learn
- PyTorch, Keras

MACHINE LEARNING & AI

- Supervised & Unsupervised Learning
- Feature Engineering
- Exploratory Data Analysis
- Deep Learning Fundamentals
- Computer Vision, NLP
- LLM APIs

TOOLS & PLATFORMS

- Git & GitHub
- Streamlit
- PowerBI
- Excel
- AWS

SOFT SKILLS

- Analytical Thinking
- Problem Solving
- Agile & Scrum
- Adaptability, Communication
- Leadership

WORK EXPERIENCE

FEB 2020 – FEB 2021

CREDENCE RESOURCE MANAGEMENT LLC, PUNE

| Customer Service Executive

- Handled inbound as well as outbound cold calls for debt collection, by keeping clear, detailed records of customer interactions, enabling accurate billing investigations.
- Maximised customer retention using exceptional communication, persuasion & relationship-building skills
- Kept client & company files safely stored on CRM system in line with confidentiality laws & regulations.

MAY 2024 – MAY 2025

SENSYCON AUTOMATION PRIVATE LIMITED, PUNE

| Trainee Project Engineer

- Assisted in writing, testing and debugging PLC ladder logic & developed HMI screens under guidance.
- Assist electricians/technicians in panel wiring, cable routing, field sensor installation. alongwith reading and understanding electrical drawings (schematics, GA drawings, panel layouts).
- Prepared and maintained technical documentation (SOPs, wiring diagrams, BOMs, FAT/SAT documents).
- Efficiently handled onsite installation and commissioning of few projects in which conducted I/O testing, dry run & trial cycles with the team. Reported issues, supported root-cause analysis, and helped implement corrections.
- Maintained professional client communication and project coordination.

JUNE 2025 – MAY 2026

PROFESSIONAL UPSKILLING & DATA SCIENCE TRANSITION